HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

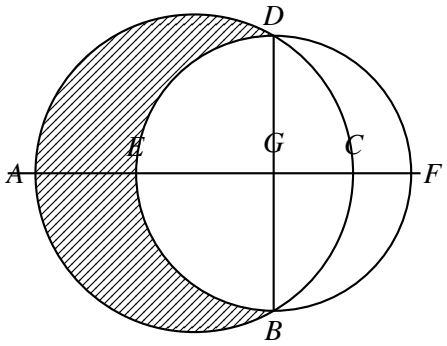
Student No.: 18 Name: _____ Marks: / 22

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In the figure, AC is a diameter of the circle $ABCD$ while BD and EF are diameters of the circle $BEDF$. It is given that C and E lie on AF. Let G be the point of intersection of AF and BD. If $AG = 30\,\text{cm}$ and $CG = 10\,\text{cm}$, find the area of the shaded region correct to the nearest cm^2.



A. $209\,\text{cm}^2$

B. $367\,\text{cm}^2$

C. $383\,\text{cm}^2$

D. $540\,\text{cm}^2$

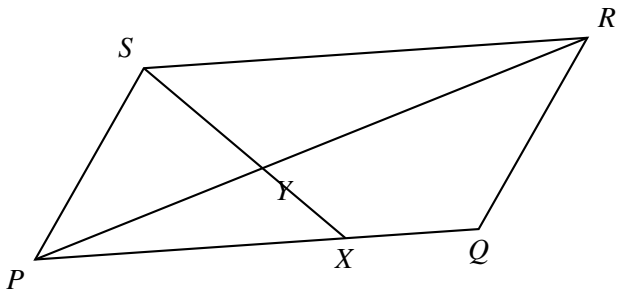
(DSE-2023 PII Q16)

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2. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,\text{cm}^2$ and $58\,\text{cm}^2$ respectively, then the area of $\triangle RSY$ is



A. $40\,\text{cm}^2$.

B. $50\,\text{cm}^2$.

C. $58\,\text{cm}^2$.

D. $72\,\text{cm}^2$.

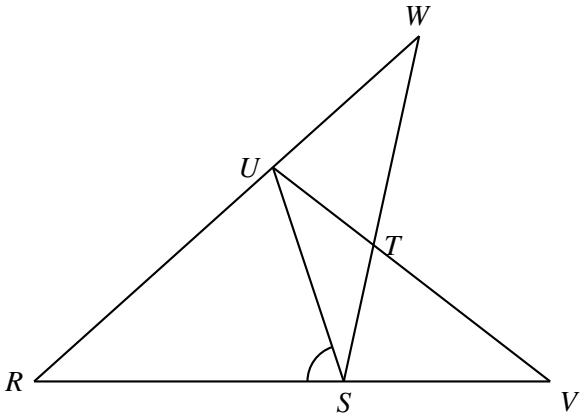
(DSE-2023 PII Q17)

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3. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. RS produced and UT produced meet at the point V while RU produced and ST produced meet at the point W. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$



A. 65°.

B. 73°.

C. 80°.

D. 82°.

(DSE-2023 PII Q22)

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4. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

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5. Let $T(n)$ be the nth term of an arithmetic sequence. If $T(11) = 83$ and $T(25) + T(30) = 463$, find the least value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) > 4 \times 10^5$.

A. 299

B. 300

C. 944

D. 945

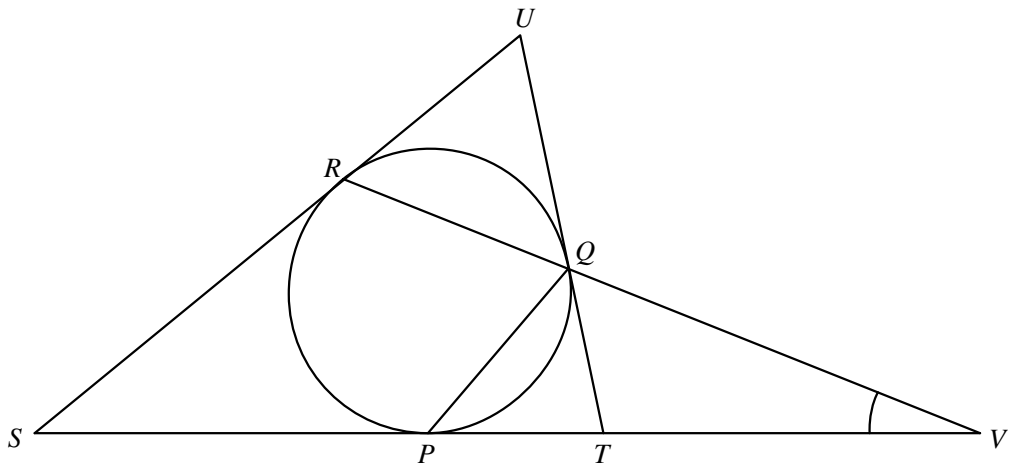
(DSE-2023 PII Q36)

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6. In the figure, P, Q and R are points lying on a circle. ST, TU and SU are the tangents to the circle at P, Q and R respectively. RQ produced and ST produced meet at the point V. If $\angle PSR = 34^\circ$ and $\angle QPT = 46^\circ$, then $\angle PVQ =$



A. 17°.

B. 22°.

C. 27°.

D. 28°.

(DSE-2023 PII Q38)

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7. The straight line $hx + ky = 6$ and the circle $x^2 + y^2 - 8x - 4y - 18 = 0$ intersect at the points M and N, where h and k are constants. If the coordinates of the mid-point of MN are $(1, 0)$, find k.

A. 4

B. 6

C. 9

D. 12

(DSE-2023 PII Q39)

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8. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. 0.

(DSE-2023 PII Q40)

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9. The equations of the straight lines L_1 and L_2 are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where k is a positive constant. It is given that L_1 cuts the x-axis at the point P. Denote the point of intersection of L_1 and L_2 by Q. If R is a point lying on L_2 such that the in-centre of $\triangle PQR$ lies on the x-axis, then the x-coordinate of R is

A. $-7k$.

B. $-k$.

C. k.

D. $7k$.

(DSE-2023 PII Q41)

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10. When a boy throws a dart, the probability that he hits the target is 0.6 . If this boy throws the dart 4 times, find the probability that he hits the target at least 2 times.

A. 0.5248

B. 0.7056

C. 0.8208

D. 0.8464

(DSE-2023 PII Q43)

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11. It is given that n is an integer. Let u, v and w be the standard deviation, the median and the range of the group of numbers $\{1-9n, 3-9n, 4-9n, 5-9n, 7-9n\}$ respectively. Which of the following must be true?

I. $u = 2$

II. $v < 4$

III. $w > 6$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q45)

I can finish _____ questions

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